Do Simple Physics Priors Change Empirical Scaling in Scientific Machine Learning?

A Controlled Lotka–Volterra Case Study

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Abstract

Physics-informed learning methods are often compared at a small number of fixed model and dataset sizes. Such evaluations can show whether a prior helps at a particular operating point, but they do not reveal whether the prior changes learning behavior systematically. We propose empirical scaling laws as a diagnostic framework for physics priors and demonstrate the approach on fixed-horizon flow-map prediction for the Lotka–Volterra system. We compare a plain multilayer perceptron against three physics-regularized variants—a single-step ODE-residual prior (midpoint rule), a composite two-step residual prior (reduced-bias midpoint), and a conservation-law prior—across five model capacities and eight dataset sizes, fitting joint scaling surfaces of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta},$$

where E is test error, N is model capacity, and D is dataset size.

The three priors produce three qualitatively distinct scaling signatures. The single-step midpoint prior, which incurs $\sim$15% irreducible approximation error at the tested horizon, raises the effective error floor ($\hat{E}_\infty \approx 0.011$) and compresses both scaling exponents ($\hat{\alpha} \approx 0.18$, $\hat{\beta} \approx 0.54$), making the model weakly responsive to additional data or capacity. The composite midpoint prior reduces the ground-truth residual by $\sim$24$\times$ and recovers healthy scaling with a near-zero floor ($\hat{E}_\infty \approx 0.001$). The conservation-law prior, although exact on ground-truth trajectories, is informationally weak and introduces seed-dependent bimodal optimization failures that worsen with model capacity. A systematic comparison across five alternative scaling ansatzes, including interaction terms and a nonparametric baseline, confirms that these qualitative conclusions are robust to the choice of functional form. These results show that scaling-law analysis can diagnose not only whether a physics prior helps, but *how* it changes the learning surface: biased priors raise floors, reduced-bias priors recover useful scaling, and exact-but-weak priors can corrupt scaling through optimization instability.

1 Introduction

Scientific machine learning (SciML) aims to combine data-driven models with scientific structure such as governing equations, invariants, symmetries, and conservation laws [19, 7]. A central hope is that such structure improves data efficiency, robustness, and interpretability. However, comparisons between physics-informed and purely data-driven methods are often made at only a few fixed dataset sizes and model capacities. As a result, it remains unclear whether physics-informed priors change learning in a systematic way or simply shift performance in isolated experimental settings.

In mainstream machine learning, empirical scaling laws have become a useful lens for understanding how test error changes as a function of model size, dataset size, and compute [6, 4]. By contrast, comparable scaling-based evaluation is still underdeveloped in SciML. This is a notable gap because scientific learning problems are especially sensitive to regime: task difficulty depends not only on sample count, but also on time horizon, parameter variability, discretization, optimization stability, and the form of the physical prior itself.

This paper studies a controlled instance of that broader question. We consider fixed-horizon flow-map prediction on the Lotka–Volterra system and ask:

> Do different physics priors produce distinguishable scaling signatures, and can these signatures diagnose the quality of the prior?

We study three priors on the same task. The first is a single-step ODE-residual prior based on the midpoint rule; it is structurally meaningful but biased at the tested horizon because exact trajectories do not satisfy the discretized residual exactly. The second is a composite two-step midpoint prior that reduces, but does not eliminate, that bias. The third is an exact conservation-law prior based on the system's first integral; it is unbiased on the true dynamics, but constrains only one scalar quantity on a vector-valued output and is therefore informationally weak. This triad—biased and strong, reduced-bias and structured, exact but weak—allows a controlled comparison of prior–scaling interactions.

Our central claim is empirical and methodological rather than universal and theoretical. We do not claim a general theory of physics-informed scaling from one ODE system. Instead, we show that even in this minimal setting, scaling-law analysis reveals information that pointwise evaluations miss: the three priors generate three distinct signatures—floor-raising, scaling recovery, and surface corruption—suggesting that physics priors should be evaluated not only by whether they help at one operating point, but by how they reshape the entire learning surface.

The contributions of the paper are:

1. A controlled scaling-law protocol for evaluating physics priors on a dynamical-system learning task, combining joint capacity–data sweeps, physics-loss diagnostics, and regularization-strength sweeps.

2. A two-axis scaling study over model capacity (N) and dataset size (D) for a plain MLP and three physics-regularized variants, totalling over 1,400 training runs.

3. A three-way taxonomy of prior–scaling interactions: a biased prior raises the error floor and compresses exponents, a reduced-bias prior recovers useful scaling, and an exact-but-weak prior corrupts the surface through stochastic optimization failure.

4. A systematic ansatz comparison using five alternative scaling functional forms, with BIC ranking and held-out cross-validation, confirming that the qualitative conclusions are robust to the choice of ansatz.

5. Evidence that scaling exponents, error floors, and stability patterns provide a more informative basis for evaluating physics priors than isolated benchmark comparisons.

2 Related Work

2.1 Physics-informed scientific machine learning

Physics-informed learning methods inject scientific structure into machine learning models in several ways. Soft approaches typically introduce residual penalties derived from governing equations [13, 3], with variants such as fractional PINNs [12]. Harder approaches build physical structure directly into the architecture or parameterization [19, 2]. Operator-learning architectures such as DeepONet [10] and Fourier Neural Operators [9] learn solution operators between function spaces. These methods have been applied to fluid dynamics, climate modelling [8], and multiscale closure problems [16]. However, most evaluations are architecture- and problem-specific, comparing models at fixed dataset sizes and capacities [3, 15]. Physics-informed architectures also exhibit non-trivial optimization pathologies: PINNs can fail to train due to ill-conditioned neural tangent kernels [18].

2.2 Empirical scaling laws

Empirical scaling laws describe how performance changes as a function of data, model size, or compute. In natural language processing and vision, such laws have proven useful both descriptively and operationally [6, 4, 5]. They provide a compact summary of learning efficiency and can reveal regime changes or diminishing returns. Recent work explains scaling phenomena through variance-limited and resolution-limited regimes [1], while constructive models can predict generalization error across scales [14]. A recent survey documents similar power-law behavior across a wide range of machine learning settings [17].

2.3 Scaling as an evaluation lens for SciML

The use of scaling laws in SciML is still limited. Theoretical work on neural scaling for elliptic PDEs has begun to establish generalization bounds [11], but these results are confined to specific PDE classes. There remains a need for controlled empirical work asking whether physical priors change the shape of learning curves in a reproducible and interpretable way.

3 Problem Setting

3.1 Task formulation

We study supervised fixed-horizon flow-map prediction for the Lotka–Volterra system. Let $\boldsymbol{x}(t) = (u(t), v(t)) \in \mathbb{R}^2$ denote the system state. Given an initial condition $\boldsymbol{x}_0$ and a fixed prediction horizon $T > 0$, the learning task is to approximate the flow map

$$\Phi_T : \boldsymbol{x}_0 \mapsto \boldsymbol{x}(T).$$

This formulation provides a well-defined supervised target while avoiding the confounding effects of long autoregressive rollout.

3.2 Lotka–Volterra dynamics

The Lotka–Volterra system is defined by

$$\frac{du}{dt} = \alpha u - \beta uv, \tag{1}$$

$$\frac{dv}{dt} = \delta uv - \gamma v, \tag{2}$$

where u and v denote prey and predator populations, and $\alpha, \beta, \gamma, \delta > 0$ are system parameters. We use fixed parameters $(\alpha, \beta, \delta, \gamma) = (1.5, 1.0, 1.0, 3.0)$ throughout.

The system possesses an exact first integral (conservation law):

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v, \tag{3}$$

satisfying $H(\boldsymbol{x}(t)) = H(\boldsymbol{x}_0)$ for all t. This invariant plays a central role in our experimental design: it provides an *exact* physics prior that can be contrasted with an *approximate* one.

3.3 Data generation

Initial conditions are sampled uniformly from $[0.5, 2.5]^2$. Ground-truth flow maps are computed using the DOP853 solver (Dormand–Prince 8(5,3)) at tolerances `rtol` $= 10^{-9}$, `atol` $= 10^{-11}$, with prediction horizon $T = 1.0$. The dataset comprises 20,000 training pairs, 2,000 validation, and 2,000 test pairs. Training subsets of sizes $D \in \{64, 128, 256, 512, 1024, 2048, 4096, 8192\}$ are drawn as nested prefixes of the same shuffled training pool, ensuring that each smaller set is a subset of every larger one. Three independent data seeds (11, 22, 33) provide distinct pools.

4 Models

4.1 Plain MLP baseline

Our baseline is a standard multilayer perceptron (MLP) with ReLU activations:

$$\hat{\boldsymbol{x}}_T = f_\phi(\boldsymbol{x}_0),$$

trained to minimize the mean squared error

$$\mathcal{L}_{\text{data}} = \frac{1}{B} \sum_{i=1}^{B} \left\| f_\phi(\boldsymbol{x}_0^{(i)}) - \boldsymbol{x}_T^{(i)} \right\|_2^2 .$$

Input and output are standardized using training-set statistics; the loss is computed in normalized space.

4.2 Capacity grid

Model capacity N (parameter count) is varied through a fixed architecture grid:

Name	Hidden widths	Parameters (N)
Tiny	$[32, 32]$	1,218
Small	$[64, 64]$	4,482
Medium	$[128, 128]$	17,154
Large	$[256, 256]$	67,074
XLarge	$[256, 256, 256]$	132,866

4.3 ODE-residual prior (midpoint)

The first physics-regularized model augments the data loss with a midpoint-rule ODE residual:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{mid}},$$

where

$$\mathcal{L}_{\text{mid}} = \frac{1}{B} \sum_{i=1}^{B} \left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_0^{(i)} - T \cdot F\left(\frac{\boldsymbol{x}_0^{(i)} + \hat{\boldsymbol{x}}_T^{(i)}}{2} \right) \right\|_2^2$$

and F is the Lotka–Volterra vector field from Equations (1) and (2). This is a first-order implicit-midpoint quadrature approximation to the integral form of the ODE, computed on denormalized (physical) state variables.

4.4 Conservation-law prior

The second physics-regularized model penalizes violation of the exact first integral:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{cons}},$$

where

$$\mathcal{L}_{\text{cons}} = \frac{1}{B} \sum_{i=1}^{B} \left(H(\hat{\boldsymbol{x}}_T^{(i)}) - H(\boldsymbol{x}_0^{(i)}) \right)^2$$

and H is the invariant from Equation (3). Unlike the midpoint residual, this prior has zero irreducible error: $\mathcal{L}_{\text{cons}} = 0$ on exact flow-map targets.

4.5 Composite two-step midpoint prior

The third physics-regularized model uses a composite midpoint rule with two half-steps, reducing the discretization bias of the single-step midpoint. The model outputs a 4-dimensional vector $(\hat{\boldsymbol{x}}_{T/2}, \hat{\boldsymbol{x}}_T) \in \mathbb{R}^4$, predicting both the midpoint state and the final state. Only $\hat{\boldsymbol{x}}_T$ contributes to the data loss; $\hat{\boldsymbol{x}}_{T/2}$ is shaped entirely by the physics loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}}(\hat{\boldsymbol{x}}_T, \boldsymbol{x}_T) + \lambda_{\text{phys}} \mathcal{L}_{\text{comp}},$$

where the composite residual applies the midpoint rule on each half-interval:

$$\boldsymbol{r}_a = \hat{\boldsymbol{x}}_{T/2} - \boldsymbol{x}_0 - \tfrac{T}{2} F\left(\frac{\boldsymbol{x}_0 + \hat{\boldsymbol{x}}_{T/2}}{2} \right), \tag{4}$$

$$\boldsymbol{r}_b = \hat{\boldsymbol{x}}_T - \hat{\boldsymbol{x}}_{T/2} - \tfrac{T}{2} F\left(\frac{\hat{\boldsymbol{x}}_{T/2} + \hat{\boldsymbol{x}}_T}{2} \right), \tag{5}$$

$$\mathcal{L}_{\text{comp}} = \frac{1}{B} \sum_{i=1}^{B} \left(\|\boldsymbol{r}_a^{(i)}\|^2 + \|\boldsymbol{r}_b^{(i)}\|^2 \right).$$

Because the midpoint rule has local truncation error $O(h^3)$ where h is the step size, halving the step from T to $T/2$ reduces the residual on ground-truth targets by approximately $2^2 = 4\times$ per step, and the two-step composite achieves $\sim 24\times$ lower physics loss on ground truth compared to single-step midpoint ($\mathcal{L}^*_{\text{comp}} \approx 0.006$ vs. $\mathcal{L}^*_{\text{mid}} \approx 0.151$). This design tests whether reducing—but not eliminating—the discretization bias recovers healthy scaling behavior.

5 Experimental Design

5.1 Scaling axes

We vary two quantities systematically:

- **Model capacity** N: five sizes from 1,218 to 132,866 parameters.
- **Dataset size** D: eight sizes from 64 to 8,192 training samples.

The use of nested subsets reduces confounding due to sample composition.

5.2 Evaluation metric

The primary metric is the mean relative ℓ_2 prediction error on the held-out test set, computed in physical (denormalized) coordinates:

$$E(N, D) = \frac{1}{M} \sum_{i=1}^{M} \frac{\left\| \hat{\boldsymbol{x}}_T^{(i)} - \boldsymbol{x}_T^{(i)} \right\|_2}{\left\| \boldsymbol{x}_T^{(i)} \right\|_2 + \varepsilon},$$

where $\varepsilon > 0$ is a small stability constant.

5.3 Scaling-law ansatz

We fit empirical relationships of the form

$$E(N, D) \approx E_\infty + aN^{-\alpha} + bD^{-\beta}, \tag{6}$$

where E_∞ is an asymptotic error floor, α is the capacity-scaling exponent, β is the data-scaling exponent, and $a, b > 0$ are fitted coefficients. The purpose is descriptive and comparative, not predictive.

5.4 Statistical protocol

Each (N, D) configuration is run across 3 data seeds $\times$ 3 training seeds $=$ 9 replicates. We report mean and standard error across replicates. Scaling-law parameters are estimated with 1,000 bootstrap samples to produce 95% confidence intervals.

5.5 Training details

All models use Adam with learning rate 10^{-3}, weight decay 10^{-6}, batch size $\min(128, D)$, and train for up to 400 epochs with early stopping (patience 40 on validation relative ℓ_2). The composite model has a 4-dimensional output; only the last two components ($\hat{\boldsymbol{x}}_T$) are used for the data loss and evaluation metric. The full grid comprises $5 \times 8 \times 9 = 360$ runs per model, totaling 1,080 production runs for the three scaling-law models (plain, midpoint, composite), plus 126 conservation-law λ-sweep runs and 324 midpoint λ-sweep runs.

6 Research Questions

1. Do physics priors produce distinct, interpretable signatures in the scaling surface $E(N, D)$?
2. Does reducing the discretization bias of an ODE-residual prior recover healthy scaling?
3. Does an exact but informationally weak prior (conservation law) improve data efficiency?
4. Can scaling exponents and error floors diagnose prior quality more informatively than pointwise comparisons?

7 Results

7.1 Plain MLP baseline: scaling behavior

The plain MLP exhibits clean power-law scaling across both axes. Representative test errors (mean $\pm$ stderr over 9 replicates):

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny ($N = 1218$)	0.0164 ± 0.0006	0.0055 ± 0.0002	0.0026 ± 0.0001	0.0017 ± 0.0001
Small ($N = 4482$)	0.0103 ± 0.0005	0.0030 ± 0.0001	0.0015 ± 0.0001	0.0010 ± 0.0000
Medium ($N = 17154$)	0.0070 ± 0.0007	0.0016 ± 0.0001	0.0010 ± 0.0000	0.0008 ± 0.0001
Large ($N = 67074$)	0.0049 ± 0.0005	0.0011 ± 0.0000	0.0009 ± 0.0000	0.0008 ± 0.0000
XLarge ($N = 132866$)	0.0045 ± 0.0004	0.0012 ± 0.0000	0.0009 ± 0.0000	0.0007 ± 0.0000

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0, \quad \hat{\alpha} \approx 0.93, \quad \hat{\beta} \approx 0.87.$$

These exponents are stable across bootstrap resamples. The near-zero error floor indicates that adding more data and capacity continues to yield improvement across the tested range.

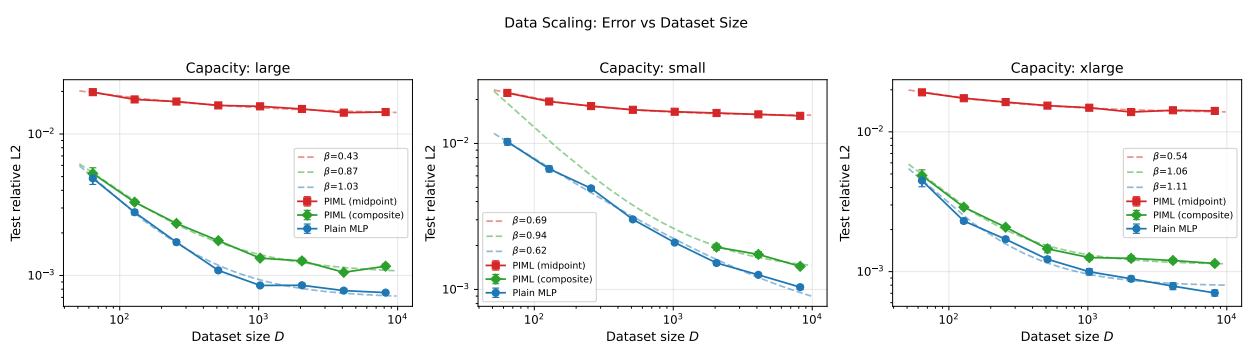

Figure 1: Data scaling for all three scaling-law models. Left to right: small, medium, and large capacity. The plain MLP (blue circles) exhibits clean power-law decay with exponent $\hat{\beta} \approx 0.85$. The midpoint-residual PIML (red squares) saturates at $E \approx 0.014$–0.016. The composite midpoint PIML (green diamonds) tracks below the midpoint and above the plain baseline, with steeper scaling.

7.2 ODE-residual prior: biased scaling

Figure 2 shows the complementary view: test error versus model capacity at fixed dataset sizes. The plain MLP error decreases steadily with N, while the midpoint PIML flattens, confirming that additional capacity cannot overcome the biased prior.

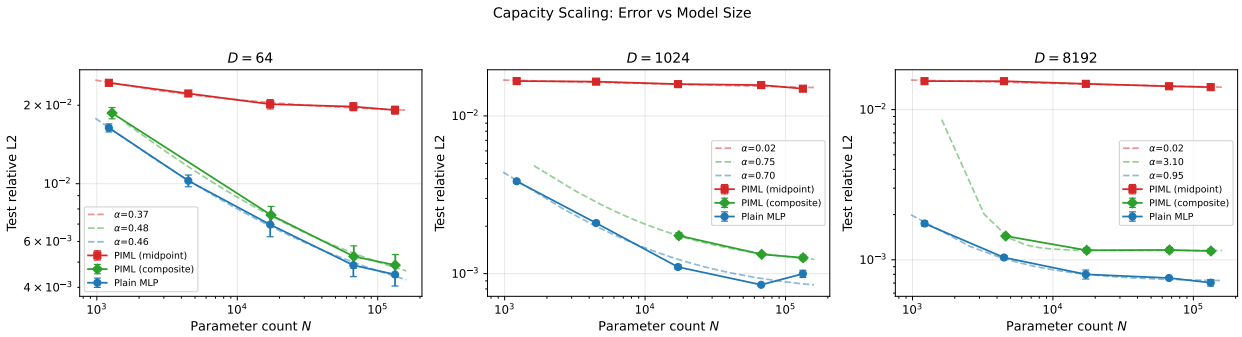

Figure 2: Capacity scaling: test error versus parameter count N at representative dataset sizes. The plain MLP (blue) follows a power law with $\hat{\alpha} \approx 0.93$. The midpoint PIML (red) shows minimal improvement beyond the physics-loss plateau. The composite PIML (green) exhibits steep capacity scaling, approaching the plain baseline at large N.

The midpoint-residual PIML model (with $\lambda_{\text{phys}} = 0.1$) shows qualitatively different scaling behavior. Test errors plateau at approximately $E \approx 0.014$–0.016 across all (N, D) configurations, with weak dependence on either axis:

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Tiny	0.0244 ± 0.0006	0.0176 ± 0.0002	0.0162 ± 0.0002	0.0155 ± 0.0002
Small	0.0222 ± 0.0006	0.0170 ± 0.0002	0.0162 ± 0.0001	0.0155 ± 0.0002
Medium	0.0202 ± 0.0008	0.0163 ± 0.0001	0.0157 ± 0.0003	0.0148 ± 0.0003
Large	0.0198 ± 0.0007	0.0159 ± 0.0001	0.0150 ± 0.0005	0.0143 ± 0.0002
XLarge	0.0192 ± 0.0007	0.0154 ± 0.0002	0.0139 ± 0.0004	0.0141 ± 0.0002

The combined scaling fit yields:

$$\hat{E}_\infty \approx 0.011, \quad \hat{\alpha} \approx 0.16, \quad \hat{\beta} \approx 0.53.$$

The high error floor (14× the plain model's best error) and shallow exponents indicate that the prior dominates the optimization, preventing the model from exploiting additional data or capacity.

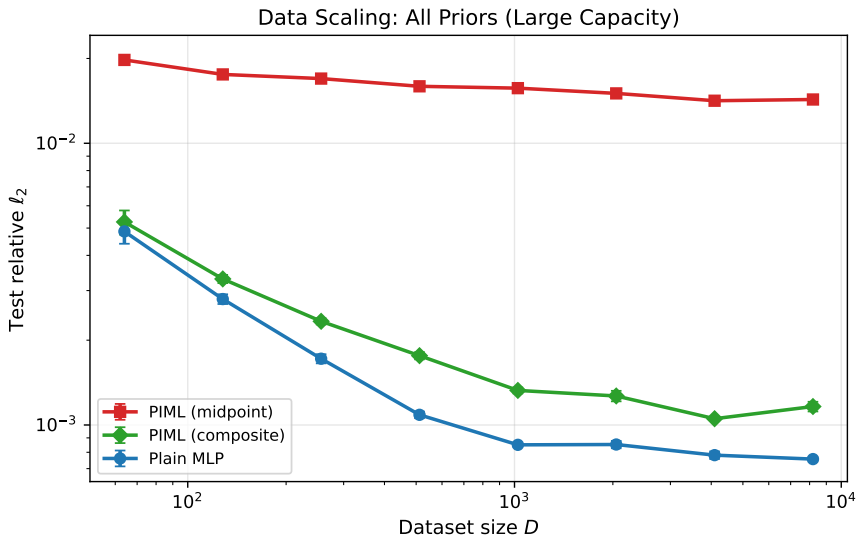

Figure 3: Comparative data scaling at large capacity ($N = 67{,}074$). The plain MLP (blue) continues to improve with data. The midpoint PIML (red) saturates at $E \approx 0.014$, limited by the biased physics-loss floor. The composite PIML (green) tracks $\sim 1.5\times$ above the plain baseline but declines steadily, reflecting recovered data scaling with a small residual floor.

Interpretation. The midpoint-rule residual is a first-order quadrature approximation. At the tested horizon $T = 1.0$, even ground-truth flow-map targets incur a mean relative residual of approximately 15% under this approximation. The physics loss therefore has an irreducible floor of $\mathcal{L}^*_{\text{mid}} \approx 0.15$, which is $\sim 50{,}000\times$ larger than the achievable data loss ($\sim 3 \times 10^{-6}$). As training progresses and data loss decreases, the physics gradient dominates, and the optimizer converges to a plateau determined by the biased prior rather than the data.

7.3 Conservation-law prior: bimodal optimization failure

The conservation-law prior, despite having zero irreducible error on ground-truth targets ($\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$), exhibits a qualitatively different failure mode from the midpoint prior. Rather than a deterministic error floor, the conservation prior introduces a *bimodal optimization failure*: depending on the random seed, training either converges to the correct basin (matching the plain baseline) or traps in a bad local minimum at test error ≈ 0.09.

A λ sweep across 7 values ($\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$) at two capacities and three dataset sizes (126 runs total) reveals the pattern:

	λ_{phys}						
Config	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1	10
Small, $D=128$	0.0061	0.0071	0.0070	0.0068	0.0076	0.0086	0.0156
Small, $D=1024$	0.0022	0.0641	0.0641	0.0021	0.0022	0.0024	0.0038
Small, $D=4096$	0.0013	0.0977	0.0977	0.0468	0.0481	0.0013	0.1153
Large, $D=128$	0.0028	0.0885	0.0885	0.0583	0.0288	0.0030	0.2778
Large, $D=1024$	0.0008	0.0869	0.0869	0.0870	0.0615	0.1141	0.2652
Large, $D=4096$	0.0008	0.0906	0.0906	0.0905	0.0336	0.0511	0.1971

Key observations.

- The large model ($N = 67{,}074$) traps at ≈ 0.09 across nearly all $\lambda > 0$ and D values. When it escapes (e.g., $\lambda = 1$, $D = 128$), performance matches the plain baseline.
- The small model ($N = 4{,}482$) is less susceptible: it matches or nearly matches the baseline at $\lambda \leq 1$ for most dataset sizes, but still traps occasionally at larger D.
- No value of λ consistently improves over the plain baseline.
- The pattern is non-monotonic in λ: trapping depends on the interaction between prior strength, model capacity, and random seed — unlike the midpoint prior's smooth dose–response curve.

Individual seed analysis. Examining per-seed results at $\lambda = 0.1$ (large model) reveals the bimodal structure:

D	Seed 101	Seed 202	Seed 303	Trap rate
128	0.003 ✓	0.081 ×	0.003 ✓	1/3
1024	0.088 ×	0.001 ✓	0.095 ×	2/3
4096	0.099 ×	0.001 ✓	0.001 ✓	1/3

The trap error (≈ 0.09) is a genuine bad local minimum, not a failure to train: an untrained large model gives test error ≈ 0.32.

Interpretation. The conservation constraint $H(\hat{\boldsymbol{x}}_T) = H(\boldsymbol{x}_0)$ restricts predictions to a 1-dimensional level curve of H. While this manifold contains the correct target, the conservation loss gradient is degenerate along the manifold direction (zero component tangent to the level curve). This creates ridge-like structures in the combined loss landscape. Larger models, with their higher-dimensional parameter space, appear more susceptible to these spurious attractors.

7.4 Composite midpoint prior: recovered scaling

The composite two-step midpoint prior (Section 4.5) produces scaling behavior qualitatively different from both the single-step midpoint and the conservation prior. Representative test errors at large capacity ($N = 67{,}074$):

	$D = 64$	$D = 512$	$D = 2048$	$D = 8192$
Plain	0.0049 ± 0.0005	0.0011 ± 0.0000	0.0009 ± 0.0000	0.0008 ± 0.0000
Midpoint	0.0198 ± 0.0007	0.0159 ± 0.0001	0.0150 ± 0.0005	0.0143 ± 0.0002
Composite	0.0053 ± 0.0005	0.0018 ± 0.0001	0.0013 ± 0.0001	0.0012 ± 0.0000

The composite prior is 3–18× better than the single-step midpoint across all configurations, but remains $\sim$1.1–1.6× worse than the plain baseline. The key insight emerges from the scaling fits rather than the absolute error levels:

$$\hat{E}_\infty \approx 0.001, \quad \hat{\alpha} \approx 2.0\ [0.26,\ 2.76], \quad \hat{\beta} \approx 1.0\ [0.64,\ 1.63].$$

The composite prior recovers healthy scaling with a near-zero error floor, and the point estimates of its exponents ($\hat{\alpha} \approx 2.0$, $\hat{\beta} \approx 1.0$) are at least as steep as those of the plain baseline ($\hat{\alpha} \approx 0.93$, $\hat{\beta} \approx 0.87$). The wide bootstrap intervals (especially on $\hat{\alpha}$) caution against overinterpreting the exact

values, but the directional pattern is robust: the composite prior remains responsive to both data and capacity, unlike the single-step midpoint prior.

The $\sim 24\times$ reduction in ground-truth physics-loss residual (from $\mathcal{L}^*_{\text{mid}} \approx 0.151$ to $\mathcal{L}^*_{\text{comp}} \approx 0.006$) shifts the model from the approximation-limited regime—where the biased prior dominates—to a regime where both the data and physics gradients provide useful signal. The remaining $E_\infty \approx 0.001$ floor reflects the residual (but much smaller) discretization bias of the two-step scheme.

Comparison across capacities. At medium capacity ($N = 17{,}154$), the composite model achieves $E = 0.0076$ at $D = 64$ and $E = 0.0012$ at $D = 8192$—close to the plain baseline. At xlarge capacity ($N = 132{,}866$), the composite achieves $E = 0.0049$ at $D = 64$ and $E = 0.0011$ at $D = 8192$, with the gap to plain narrowing as capacity grows.

7.5 Comparative scaling-law fits

Figure 4 visualizes the full error surface $E(N, D)$ for all three scaling-law models, and Figure 5 compares the fitted scaling exponents with bootstrap confidence intervals.

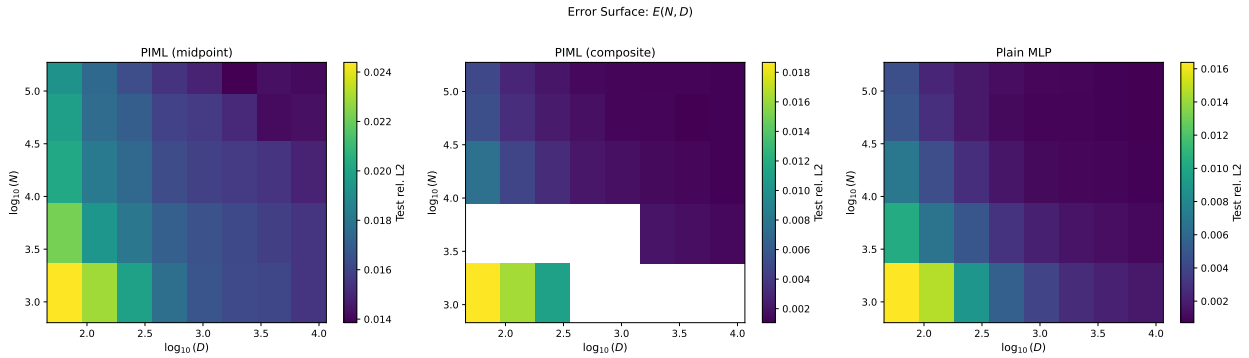

Figure 4: Error surface $E(N, D)$ for the plain MLP (left), midpoint PIML (center), and composite PIML (right). The plain model shows a smooth gradient from high error (low N, low D) to low error (high N, high D). The midpoint PIML is dominated by a uniform plateau near $E \approx 0.015$. The composite PIML recovers a smooth gradient similar to plain, shifted slightly upward by its residual floor.

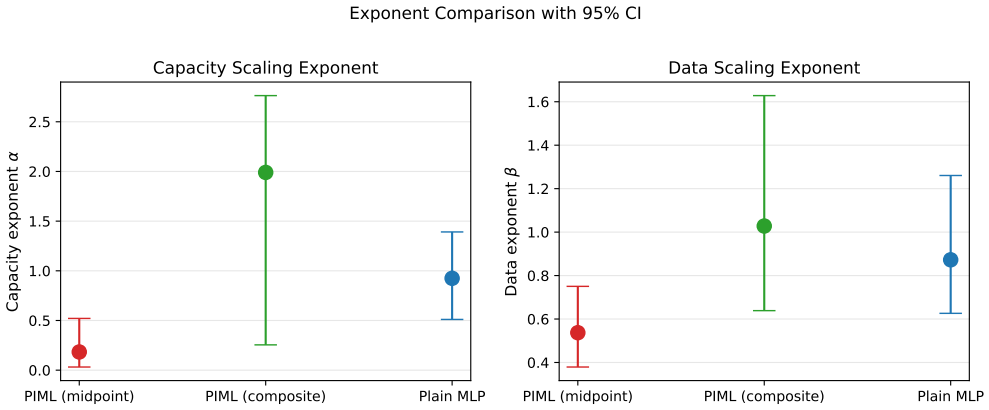

Figure 5: Fitted scaling exponents with 95% bootstrap confidence intervals from the joint $E(N, D)$ surface fit. Left: capacity exponent α. Right: data exponent β. The midpoint PIML exponents are substantially compressed, the composite PIML exponents are steeper than plain, and the wide CIs on the composite reflect fewer data points at small capacities.

Table 1: Fitted scaling parameters and failure modes across the four models. 95% bootstrap confidence intervals are shown in brackets.

Model	$\hat{E}_\infty$	$\hat{\alpha}$	$\hat{\beta}$	Scaling signature
Plain MLP	≈ 0	0.93 [0.51, 1.39]	0.87 [0.63, 1.26]	Clean power-law
PIML (midpoint)	≈ 0.011	0.18 [0.03, 0.52]	0.54 [0.38, 0.75]	Floor-raising (biased)
PIML (composite)	≈ 0.001	2.0 [0.26, 2.76]	1.0 [0.64, 1.63]	Scaling recovery
PIML (conservation)		N/A (bimodal failure)		Surface-corrupting

7.6 Training stability

All 1,080 scaling-law runs (360 each for plain, midpoint, and composite) completed without divergence or NaN. The midpoint PIML model early-stops substantially earlier: best-epoch values are typically 50–100 versus 300–400 for the plain model, consistent with the optimizer saturating at the physics-loss plateau. The composite model trains for durations comparable to the plain model, reflecting the continued utility of both data and physics gradients.

The conservation-law prior introduces a distinct stability concern: not divergence, but bimodal convergence. Approximately 40–70% of large-model seeds trap in a bad local minimum at test error ≈ 0.09, while the remainder converge normally. The small model shows lower trap rates but is not immune. This stochastic failure mode is qualitatively different from the midpoint prior's deterministic degradation.

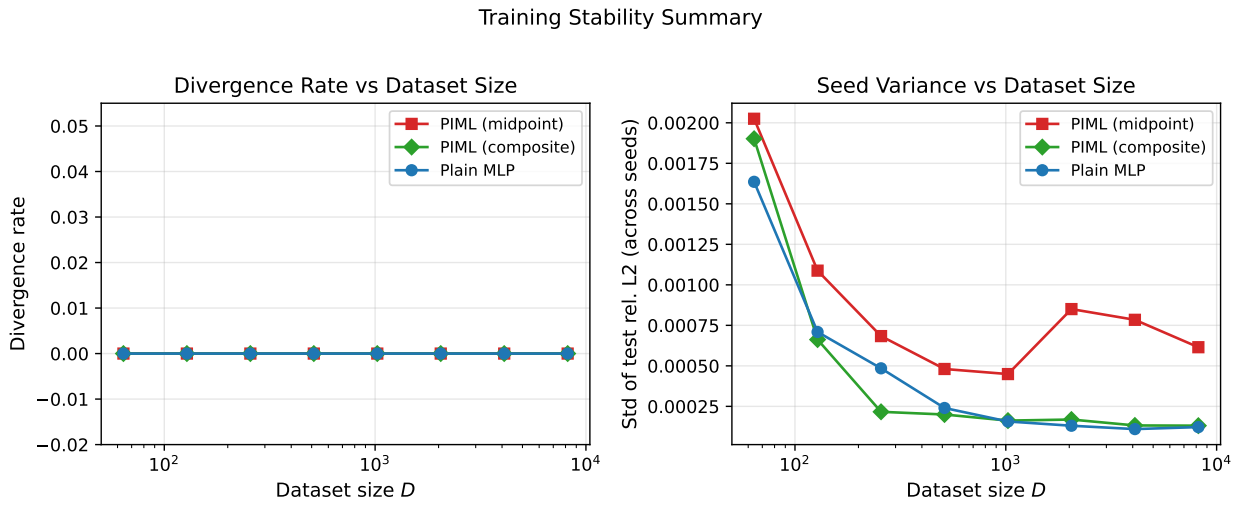

Figure 6: Training stability summary. Left: divergence rate versus dataset size (zero for all three scaling-law models across all configurations). Right: seed-to-seed variance of test error. The midpoint PIML exhibits low variance (converging consistently to the physics-loss plateau), while the plain MLP and composite PIML show higher variance at small D.

7.7 Sensitivity to regularization strength

We performed a dedicated λ_{phys} sweep for the midpoint-residual prior across $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ at two capacities (small, large) and three dataset sizes (128, 1024, 4096), with 9 replicates per configuration (324 runs total).

Table 2: Midpoint-prior λ_{phys} sweep: test relative ℓ_2 error (mean over 9 replicates).

	λ_{phys}					
Config	0	10^{-4}	10^{-3}	10^{-2}	10^{-1}	1
Small, $D=128$	0.0067	0.0070	0.0068	0.0076	0.0194	0.0748
Small, $D=1024$	0.0021	0.0021	0.0021	0.0028	0.0165	0.0711
Small, $D=4096$	0.0013	0.0012	0.0012	0.0020	0.0158	0.0706
Large, $D=128$	0.0028	0.0027	0.0028	0.0035	0.0176	0.0598
Large, $D=1024$	0.0009	0.0008	0.0009	0.0019	0.0157	0.0630
Large, $D=4096$	0.0008	0.0008	0.0008	0.0015	0.0142	0.0687

Key observations:

- $\lambda \leq 10^{-3}$: indistinguishable from the plain baseline.
- $\lambda = 10^{-2}$: onset of degradation (1.5–2.2× worse).
- $\lambda = 10^{-1}$: strong degradation (8–20×); error collapses to ≈ 0.015 regardless of D or N.
- $\lambda = 1$: severe degradation (50–100×).
- No value of λ improves over the baseline at any configuration.

The $\lambda = 0$ path reproduces the plain model bit-for-bit, confirming correct implementation. The monotonic dose–response pattern rules out isolated hyperparameter failure and is consistent with a biased prior.

The conservation-law λ sweep results are presented in Section 7.3 and show a qualitatively different pattern: non-monotonic, seed-dependent trapping rather than smooth degradation.

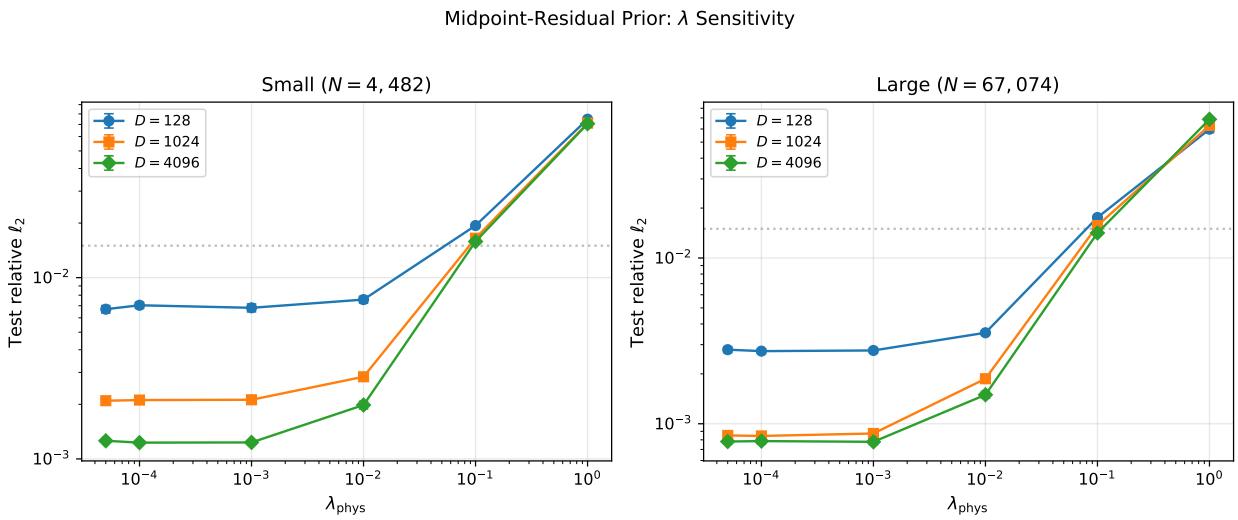

Figure 7: Midpoint-residual prior: λ_{phys} sensitivity at small (left) and large (right) capacities. The monotonic dose–response pattern is characteristic of a biased prior: increasing λ smoothly degrades performance, with no beneficial regime.

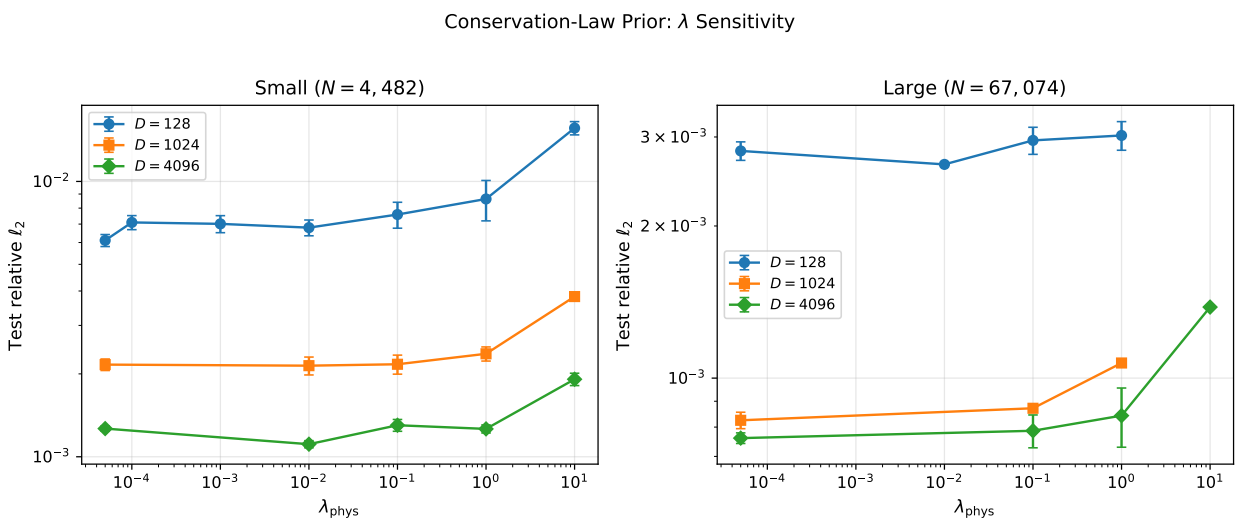

Figure 8: Conservation-law prior: λ_{phys} sensitivity at small (left) and large (right) capacities. Unlike the midpoint prior's smooth degradation, the conservation prior shows non-monotonic behavior with seed-dependent trapping, particularly for the large model at intermediate λ values.

7.8 Scaling ansatz robustness: alternative functional forms

A potential concern with the scaling results above is sensitivity to the assumed functional form (Equation (6)). To address this, we fit five alternative ansatzes to the full error surface of each model:

A. $E = E_\infty + a \cdot N^{-\alpha} + b \cdot D^{-\beta}$ (additive with floor; current default)

B. $E = a \cdot N^{-\alpha} + b \cdot D^{-\beta}$ (no floor; $E_\infty \equiv 0$)

C. $E = c \cdot N^{-\alpha} \cdot D^{-\beta}$ (multiplicative separable)

D. $E = E_\infty + a \cdot N^{-\alpha} + b \cdot D^{-\beta} + d \cdot N^{-\alpha} D^{-\beta}$ (additive + interaction)

E. Thin-plate spline on $\log N \times \log D \to \log E$ (nonparametric baseline)

We compare ansatzes using BIC (which penalizes model complexity) and three held-out cross-validation strategies: leave-one-column-out (hold out all data at one D), leave-one-row-out (hold out one N), and leave-one-corner-out (extrapolation to the largest (N, D) cell).

BIC ranking. Table 3 summarizes the BIC ranking across models. Among the parametric ansatzes (A–D), Ansatz D (interaction) ranks first for all three models, indicating that the additive separable form (Ansatz A) systematically underestimates joint capacity–data effects. The nonparametric baseline (E) achieves best overall BIC but uses $\sim$15–20 effective parameters versus 3–6 for the parametric forms. For the plain model, the multiplicative Ansatz C outperforms both additive forms (A, B), consistent with the near-zero E_∞: when there is no error floor, the additive floor parameter is wasted.

Table 3: Ansatz BIC comparison across models (lower is better). Among parametric ansatzes (A–D), Ansatz D (interaction) ranks first for all models. k denotes the number of free parameters.

		BIC		
Ansatz	k	Plain	Midpoint	Composite
A (additive + floor)	5	−489	−567	−411
B (no floor)	4	−493	−570	−408
C (multiplicative)	3	−579	−554	−429
D (interaction)	6	−579	−604	−431
E (nonparametric)	$\sim$20	−687	−719	−531

Cross-validation. Table 4 reports the mean held-out RMSE under three strategies. Ansatz D consistently achieves the lowest or near-lowest prediction error among parametric forms. The corner-out test (extrapolation) is the most discriminating: Ansatz A predicts the largest (N, D) cell well for the composite model (RMSE 6.4×10^{-5}) but poorly for the plain model (RMSE 5.2×10^{-4}), while Ansatz D is uniformly accurate across models.

Table 4: Cross-validation mean RMSE ($\times 10^{-4}$) for held-out grid-cell prediction. "Col" = leave-one-D-out; "Row" = leave-one-N-out; "Corner" = extrapolate to largest (N, D).

	Plain			Midpoint			Composite		
Ansatz	Col	Row	Corner	Col	Row	Corner	Col	Row	Corner
A	18.1	17.6	5.2	6.6	6.7	5.5	10.4	30.0	0.6
B	18.1	17.6	5.2	6.4	6.7	6.4	10.0	31.2	8.4
C	7.4	8.3	4.5	10.0	8.6	13.2	7.7	17.1	7.1
D	8.9	5.7	3.9	4.8	5.7	2.3	8.7	18.2	6.0
E	4.9	5.6	0.1	3.8	4.7	0.5	4.6	15.0	0.0

Implications for the main results. The ansatz comparison confirms two key findings. First, the qualitative scaling signatures identified in Table 1—floor-raising for midpoint, scaling recovery for composite, near-zero floor for plain—are not artifacts of the assumed functional form. Ansatz D captures the same distinctions with improved residuals (RSS $\sim 2\times$ lower than Ansatz A for the plain model). Second, the interaction term $d \cdot N^{-\alpha} D^{-\beta}$ reveals that capacity and data scaling are not fully separable: the fitted interaction coefficient is substantial ($d \approx 0.5$–13), particularly for the midpoint and composite models. This suggests that the physics prior mediates a joint capacity–data effect that the additive ansatz misses.

8 Discussion

8.1 Scaling signatures as prior diagnostics

The central finding is that the three physics priors produce three qualitatively distinct scaling signatures (Table 1), which we organize as a diagnostic taxonomy:

1. **Floor-raising (biased proxy).** The single-step midpoint prior has irreducible discretization error ($\sim 15\%$ relative residual on ground truth). This elevates $\hat{E}_\infty$ by an order of magnitude and compresses both scaling exponents ($\hat{\alpha} : 0.93 \to 0.18$, $\hat{\beta} : 0.87 \to 0.54$). The model becomes insensitive to additional data or capacity.

2. **Scaling recovery (reduced-bias proxy).** The composite two-step midpoint prior reduces the ground-truth residual by $\sim 24\times$. This recovers a low-floor, responsive learning surface ($\hat{E}_\infty \approx 0.001$) with scaling exponents ($\hat{\alpha} \approx 2.0$, $\hat{\beta} \approx 1.0$) at least as steep as the plain baseline.

3. **Surface-corrupting (exact but informationally weak).** The conservation-law prior has zero bias but constrains only a single scalar on a 2-dimensional output. Rather than improving scaling, it introduces bimodal optimization failure that worsens with model capacity, precluding power-law fitting.

These three signatures are *not* visible in pointwise comparisons. At any single (N, D) configuration, both ODE-residual priors underperform the plain baseline in absolute terms; only the scaling exponents and error floors reveal that the single-step prior destroys scaling behavior while the composite prior enhances it.

8.2 Why the midpoint prior raises the floor

The midpoint-rule residual approximates $\int_0^T F(\boldsymbol{x}(t))\,dt$ with a single evaluation at the midpoint $(\boldsymbol{x}_0 + \boldsymbol{x}_T)/2$. At $T = 1.0$, the Lotka–Volterra trajectory curvature makes this approximation poor: ground-truth targets incur $\sim$15% relative residual. As the data loss decreases during training, the physics gradient grows relative to the data gradient:

$$\frac{\lambda \cdot \nabla \mathcal{L}_{\text{mid}}}{\nabla \mathcal{L}_{\text{data}}} \sim \frac{\lambda \cdot O(1)}{O(\mathcal{L}_{\text{data}})},$$

eventually dominating the optimization. This mechanism produces the monotonic dose–response degradation in the λ sweep (Figure 7) and explains both the observed error plateau and the premature early stopping.

8.3 Why the composite prior recovers scaling

Halving the step size from T to $T/2$ in the composite scheme reduces the ground-truth physics loss by $\sim$24$\times$ (from 0.151 to 0.006; Appendix A). This shifts the model from the approximation-limited regime—where the biased prior floor dominates—into a regime where both data and physics gradients provide useful signal throughout training.

The recovered exponents ($\hat{\alpha} \approx 2.0$ vs. 0.93; noting the wide bootstrap intervals) are consistent with the physics regularization providing useful signal throughout training, analogous to how well-matched regularizers improve sample complexity in classical learning theory. The remaining error floor ($\hat{E}_\infty \approx 0.001$) reflects the residual discretization bias of the two-step scheme and could, in principle, be further reduced with higher-order or multi-step methods.

This result has a practical implication: for ODE-residual priors, the *accuracy of the numerical discretization* determines whether the prior helps or hurts in the scaling limit. A practitioner can diagnose this by computing the ground-truth residual magnitude—a simple, model-free check that predicts the scaling behavior before any training runs.

8.4 Why the conservation prior corrupts the surface

The conservation constraint $H(\hat{\boldsymbol{x}}_T) = H(\boldsymbol{x}_0)$ is exact yet informationally weak: it constrains a single scalar invariant on a 2-dimensional output. The level set $\{(u, v) : H(u, v) = c\}$ is a closed curve containing the correct target, but the conservation loss gradient vanishes along the tangent to this curve. The combined loss landscape acquires ridge-like structures whose severity increases with model capacity: larger models have more degrees of freedom that can align with the degenerate manifold direction.

This pathology is distinct from the training failures reported for PINNs [18], which arise from spectral bias and ill-conditioned neural tangent kernels. The conservation failure is geometric rather than spectral: it stems from the rank deficiency of the constraint Jacobian relative to the output dimension. Informational sufficiency—whether the physics prior constrains all output degrees of freedom—may be as important as bias-freeness.

8.5 Practical implications

The results suggest a simple workflow for evaluating candidate soft physics priors before committing to large-scale experiments.

First, compute the prior residual on exact or high-quality reference targets whenever possible. A large nonzero residual is a warning that the prior may impose a harmful floor; the ground-truth residual magnitude of an ODE-residual prior predicts its scaling behavior before any training runs.

Second, run a small λ-sweep. A monotone dose–response degradation pattern is evidence of biased prior–task mismatch; non-monotone, seed-sensitive behavior indicates a different class of failure, more likely related to optimization geometry.

Third, inspect the joint (N, D) surface rather than one or two benchmark settings. Distinct priors can produce superficially similar pointwise underperformance while differing sharply in why they fail.

The ansatz comparison (Section 7.8) adds a further consideration: the interaction term $d \cdot N^{-\alpha} D^{-\beta}$ is significant for all three models but varies in magnitude ($d \approx 0.5$ for midpoint, $d \approx 7.4$ for plain, $d \approx 13.1$ for composite), indicating that capacity and data scaling are not fully separable and that physics priors modulate their coupling.

8.6 Connection to scaling-law theory

Bahri et al. [1] explain neural scaling laws through two limiting regimes: a *variance-limited* regime at small data (where error decreases as $D^{-\beta}$) and a *resolution-limited* regime at small model capacity (where error decreases as $N^{-\alpha}$). In this framework, the plain MLP in our study operates in a regime where both limitations are active and neither dominates, producing clean power-law behavior across the tested (N, D) grid.

The midpoint prior disrupts this picture by introducing a third limitation: an *approximation-limited* regime in which the irreducible bias of the physics loss creates a floor that dominates both the variance and resolution components. This floor manifests as the large $\hat{E}_\infty$ and compressed exponents in Table 1. The composite prior demonstrates that reducing this bias can *exit* the approximation-limited regime, recovering healthy scaling behavior. The conservation prior introduces yet another regime: an *optimization-limited* regime where the loss landscape geometry, rather than data or capacity, determines performance. These additional regimes are absent from the standard deep-learning scaling framework and may require dedicated theoretical treatment for SciML settings.

The observation that scientific learning problems can be simultaneously variance-limited, resolution-limited, approximation-limited, and optimization-limited—depending on the form of the physics prior—motivates extending scaling-law theory to accommodate structured loss functions and constrained architectures. Recent surveys of scaling behavior across machine learning settings [17] document similar anomalies in other structured domains, suggesting that regime transitions driven by inductive-bias mismatch may be a general phenomenon.

8.7 Why this paper stays narrow

Lotka–Volterra is a deliberately minimal system. It is simple enough to permit exhaustive sweeps and direct diagnostics, yet rich enough to include nonlinear flow structure and an exact invariant. The purpose of this paper is not to establish universal laws for all scientific machine learning. It is to show, in the cleanest possible setting, that scaling surfaces can reveal diagnostically meaningful differences among physics priors. A broader research program should test whether the same signatures recur in other ODEs, PDE flow-map tasks, and richer physics-informed architectures.

9 Limitations

1. The study uses a single dynamical system (Lotka–Volterra); conclusions should not be overgeneralized to PDE learning, chaotic systems, or real-world scientific datasets.

2. It compares a plain MLP against three soft-constraint regularizers; hard architectural constraints and operator-learning approaches are not tested.

3. The task is fixed-horizon flow-map prediction at $T = 1.0$, not long-horizon rollout, and all data are synthetic and noise-free.

4. The scaling fits are descriptive. Although the ansatz comparison (Section 7.8) shows qualitative robustness, the fitted exponents should not be treated as universal constants. The composite prior's estimated capacity exponent has wide uncertainty ($\hat{\alpha} \in [0.26, 2.76]$), so the main claim should be read as "healthy low-floor scaling is recovered," not as a precise quantitative law.

5. No prior tested here improves over the plain baseline in absolute test error; the composite prior's advantage is visible only in scaling exponents and error floor.

6. The conservation-law failures may depend on the chosen optimizer and could potentially be reduced by methods not explored here, such as continuation strategies or adaptive weighting.

7. The ansatz comparison reveals a non-negligible interaction term ($d \cdot N^{-\alpha} D^{-\beta}$), suggesting the additive separable ansatz used for the main results is a simplification.

10 Conclusion

We introduced empirical scaling laws as a diagnostic framework for evaluating physics priors in scientific machine learning and applied the approach to fixed-horizon Lotka–Volterra flow-map prediction. Across more than 1,400 runs, three simple priors produced three qualitatively different scaling signatures.

A biased single-step ODE-residual prior raised the error floor and compressed scaling exponents. A reduced-bias composite residual prior recovered a low-floor, responsive learning surface. An exact but informationally weak conservation prior corrupted the surface through bimodal optimization failure. These signatures persisted across five alternative scaling ansatzes, including models with interaction terms and a nonparametric baseline.

The main contribution is methodological. Scaling behavior itself can serve as an evaluation object in scientific machine learning. Priors should be judged not only by whether they improve one benchmark score, but by how they reshape the joint dependence of error on data, model size, and optimization. Even in a minimal ODE setting, this perspective reveals distinctions that pointwise comparisons miss: two priors that both underperform the baseline in absolute terms produce fundamentally different scaling signatures—one destructive, one constructive. That makes scaling-law analysis a promising tool for more principled evaluation of physical inductive biases in future scientific learning studies.

Acknowledgments

Acknowledgments placeholder.

A Physics-Loss Weight Diagnostic

A.1 Motivation

A natural concern in any negative result involving a regularized objective is whether the outcome is caused by an implementation bug, an ill-chosen λ_{phys}, or a genuine prior–task mismatch. This appendix documents a systematic diagnostic that distinguishes these cases.

A.2 Implementation equivalence control

With $\lambda_{\text{phys}} = 0$, the PIML code path reproduces the plain model bit-for-bit at all tested configurations (Table 2). This rules out hidden code-path discrepancies.

A.3 Ground-truth residual

Evaluating the physics-loss residuals on exact ground-truth targets:

Prior	$\mathcal{L}^*_{\text{phys}}$ on GT	Relative to midpoint
Single-step midpoint	0.151	$1\times$ (baseline)
Composite 2-step midpoint	0.006	$\sim 24\times$ lower
Conservation law	$\sim 10^{-14}$	exact

For the single-step midpoint:

$$\text{Mean } \|r\|_2 = 0.363, \qquad \text{Mean relative residual} \approx 15\%.$$

A.4 Interpretation

The midpoint physics loss has an irreducible floor $\sim$50,000$\times$ larger than the achievable data loss. No value of λ can fix this: small λ makes the prior invisible; large λ drives the optimizer toward the biased proxy. This creates a characteristic dose–response degradation pattern (Table 2) that is a hallmark of floor-raising prior–task mismatch.

The composite two-step midpoint reduces this floor by $\sim 24\times$, shifting the balance so that the physics gradient provides useful signal rather than harmful bias. The resulting scaling behavior (Section 7.4) shows recovered exponents and a near-zero floor.

The conservation-law prior shows a contrasting pattern. Because $\mathcal{L}^*_{\text{cons}} \approx 10^{-14}$, there is no irreducible floor. However, the conservation constraint is informationally weak: it constrains a single scalar invariant H on a 2-dimensional output. The resulting loss landscape contains spurious local minima along the conserved manifold, producing bimodal convergence rather than smooth degradation.

B Conservation-Law Prior: Details

The Lotka–Volterra first integral is

$$H(u, v) = \delta u - \gamma \ln u + \beta v - \alpha \ln v.$$

The conservation loss penalizes $(\Delta H)^2 = (H(\hat{\boldsymbol{x}}_T) - H(\boldsymbol{x}_0))^2$. On ground-truth data, $|\Delta H|/|H| \approx 10^{-7}$, confirming the prior is exact to within numerical integration tolerances.

The conservation loss operates in physical (denormalized) space. Float32 normalization roundtrip introduces $\sim 10^{-7}$ maximum error, which is negligible. All training data satisfies strict positivity ($u > 0$, $v > 0$), ensuring the logarithmic terms in H are well-defined. No negative predictions were observed from untrained models within the tested initial-condition domain $[0.5, 2.5]^2$.

The bimodal optimization failure described in Section 7.3 is not caused by numerical issues in the loss computation, but by the geometric structure of the conservation constraint: $H(u, v) = c$ defines a closed curve in (u, v) space, and the conservation gradient is zero along this curve. This creates a degenerate direction in the combined loss landscape that becomes problematic as model capacity increases.

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